

Complete DOM (Document Object Model) Guide

The DOM (Document Object Model) is a programming interface for HTML and XML documents. It allows JavaScript to dynamically access, manipulate, create, update, and delete webpage elements.

1. What is DOM?

When a browser loads an HTML page, it converts the HTML into a tree-like structure called the DOM Tree.

2. Why DOM Exists

DOM allows JavaScript to change HTML, change CSS, react to user actions, add new elements, remove elements, and build interactive webpages.

3. Types of Nodes

- Document Node
 - Element Node
 - Text Node
 - Attribute Node
 - Comment Node
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4. Window vs Document Object

Window represents the browser window.
Document represents the webpage document.

5. DOM Tree Relationships

Parent, Child, and Sibling relationships exist between DOM elements.

6. Selecting Elements

Methods:

- getElementById()
 - getElementsByClassName()
 - getElementsByTagName()
 - querySelector()
 - querySelectorAll()
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7. NodeList vs HTMLCollection

HTMLCollection is live.
NodeList is usually static.
NodeList supports `forEach()`.

8. Changing Content

`innerHTML` → changes HTML.
`innerText` → visible text.
`textContent` → all text.

9. Changing CSS

Use style property: `element.style.color = "red";`

10. Attributes

`getAttribute()`
`setAttribute()`
`removeAttribute()`

11. Creating Elements

Use `createElement()` and `appendChild()`.

12. Removing Elements

Use `remove()` or `removeChild()`.

13. Event Handling

Events include click, submit, keydown, mouseover, etc. Use `addEventListener()`.

14. Event Bubbling & Capturing

Bubbling moves upward.
Capturing moves downward.

15. Event Delegation

Using parent elements to handle child events efficiently.

16. DOM Traversal

parentElement, children, nextElementSibling, previousElementSibling, firstElementChild.

17. Forms in DOM

Access forms using document.forms and input values using value property.

18. Class Manipulation

```
classList.add()  
classList.remove()  
classList.toggle()  
classList.contains()
```

19. Timing Events

```
setTimeout()  
setInterval()
```

20. Loading Events

DOMContentLoaded and load events.

21. BOM vs DOM

BOM controls browser.
DOM controls webpage.

22. Important DOM Properties

innerHTML, innerText, textContent, style, value, checked, src, href.

23. Important DOM Methods

getElementById(), querySelector(), createElement(), appendChild(), remove().

24. MutationObserver

Used to detect DOM changes dynamically.

25. Virtual DOM

Used by React for efficient rendering.

26. Shadow DOM

Creates isolated DOM trees in Web Components.

27. Rendering Process

HTML → DOM → CSSOM → Render Tree → Paint.

28. Performance Optimization

Avoid excessive DOM manipulation. Use DocumentFragment for efficiency.

29. Interview Topics

- Event Delegation
 - Bubbling vs Capturing
 - NodeList vs HTMLCollection
 - innerHTML vs innerText
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30. Real-Life Uses

Forms, animations, games, chat apps, React apps, e-commerce websites.

Example DOM Code

```
const btn = document.getElementById("btn");  
btn.addEventListener("click", function() {  
  alert("Button Clicked");  
});
```

THOUGHT: Mastering DOM means mastering how websites become interactive.

“Learning gives creativity, creativity leads to thinking.” — Dr. A. P. J. Abdul Kalam